

GENERATIVE AI

From Ideas to Intelligent Creation

- What it is
- How it works
- Applications
- Future impact

What is Generative AI?

AI that creates new content from learned patterns.

- Text • Images • Audio • Video • Code
- Powered by foundation models
- Responds to natural-language prompts

How Generative AI Works

Prompt → Model → Generated Output

- Training on large datasets
- Learning patterns and relationships
- Generating new content probabilistically
- Human feedback improves quality

Generative AI vs Traditional AI

A shift from prediction to creation

- Traditional AI: classifies, predicts, detects
- Generative AI: creates, transforms, summarizes
- Example: spam detection vs email generation

Large Language Models (LLMs)

The engine behind modern AI assistants

- Tokens represent pieces of text
- Transformers learn context
- Attention connects related information
- Examples: chatbots, coding assistants, document analysis

Prompt Engineering

Better instructions → Better results

- Be specific about the goal
- Provide context and constraints
- Give examples when useful
- Ask for a structured output
- Iterate and refine

Real-World Applications

Generative AI is becoming a co-pilot across industries

- Software: code generation & debugging
- Education: tutoring & content creation
- Business: reports, research & automation
- Healthcare: documentation & decision support
- Creative: images, video & design

Agentic AI

From generating answers to taking actions

- Perceive → Reason → Plan → Act
- Uses tools, APIs and external systems
- Maintains context and executes workflows
- Human oversight remains important

Challenges & Responsible AI

Capability must be matched with responsibility

- Hallucinations and inaccurate outputs
- Bias and fairness
- Privacy and data security
- Copyright and intellectual property
- Human verification for high-impact decisions

The Future of Generative AI

AI will increasingly work alongside people

- Multimodal and more capable models
- Smarter AI agents and automation
- Personalized learning and productivity
- Human + AI collaboration
- Key skill: knowing how to direct and verify AI